2 Figure 2.1 shows a ship made of steel floating on the sea.

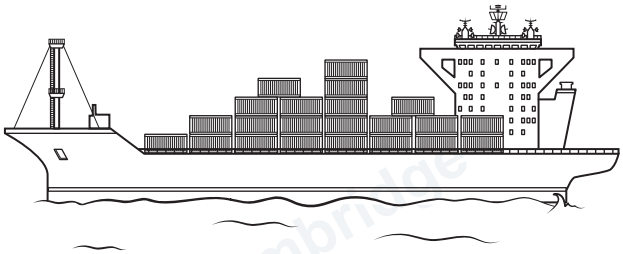


Figure 2.1

(a) (i) State the equation that links density, mass and volume.

..... [1]

(ii) The density of steel is 7900 kg / m^3. The mass of a solid steel cube is 110 kg.

Calculate the volume of the steel cube.

volume = ... m^3 [1]

(iii) The density of sea water is 1030 kg / m^3.

State why the solid steel cube sinks when placed in sea water.

.....

..... [1]

(iv) Explain why the steel ship in Figure 2.1 floats on the sea water.

.....

.....

..... [2]

DO NOT WRITE IN THIS MARGIN

© Cambridge University Press & Assessment 2026

(b) The mass of the ship is 3.5×10^7 kg. The ship accelerates at $0.75\,m/s^2$.

(i) Calculate the force which causes this acceleration.

force = ... N [2]

(ii) Explain why the force provided by the ship's engine must be larger than the value in **2(b)(i)**.

.....

..... [2]

[Total: 9]

DO NOT WRITE IN THIS MARGIN

© Cambridge University Press & Assessment 2026